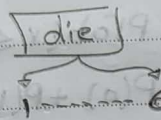
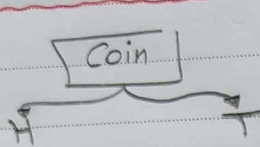


## → Random Experiment



Sample Space: set of all Possible outcomes

$$\Omega = \{ 1, 2, 3, 4, 5, 6 \}$$

→ Omega      → subset

$$0 \leq \text{Prob. (event)} \leq 1$$

## Random Variable

### Discrete

discrete random variable  $x$  has a countable number of possible values

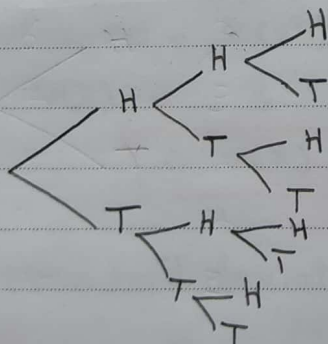
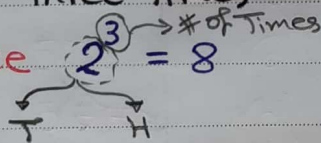
### Continuous

continuous random variable  $x$  takes all values in a given interval of numbers

## Discrete Random Number

Ex Tossing a fair coin Three Times

∴ # of sample space  
→ number



H H H  
H H T  
H T H  
H T T  
T H H  
T H T  
T T H  
T T T

all possible outcomes  
(Sample Space)

$x \rightarrow$  number of head

$x = 0, 1, 2, 3$

→ Prob

| x        | f | Prob(x)       |
|----------|---|---------------|
| 0        | 1 | $\frac{1}{8}$ |
| 1        | 3 | $\frac{3}{8}$ |
| 2        | 3 | $\frac{3}{8}$ |
| 3        | 1 | $\frac{1}{8}$ |
| $\Sigma$ | 8 | 1             |

→ cdf

$$F(2) = \text{Prob}(x \leq 2)$$

$$F(2) = P(0) + P(1) + P(2)$$

$$F(2) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}$$

## → Probability density function (PDF)

$$f(x) = P(a) = \text{Prob.}(x=a)$$

$$* 0 \leq P(x) \leq 1$$

$$* \sum_{i=1}^n P(x_i) = 1$$

It must happen in order for the  $f(x)$  to be PDF

## → Cumulative distribution function (CDF)

$$F(x) = \sum_{x_i \leq a} P(x_i)$$

$$F(x) = \text{Prob.}(x \leq a)$$

## → Expected value (Mean)

$$E(x) = \mu_x = \sum_{i=1}^n x_i P(x_i)$$



## → Variance

$$\begin{aligned}
 V(x) &= \sigma^2 = E((x - E(x))^2) \\
 &= E(x^2) - (E(x))^2 \\
 &= \sum_{i=1}^n x_i^2 P(x_i) - \left( \sum_{i=1}^n x_i P(x_i) \right)^2
 \end{aligned}$$

## → Standard deviation

$$SD = \sigma = \sqrt{\text{variance}}$$

← *freq*

| $x$      | $f$ | $P(x)$        | $x P(x)$      | $x^2 P(x)$     | $F(x)$        |
|----------|-----|---------------|---------------|----------------|---------------|
| 0        | 1   | $\frac{1}{8}$ | 0             | 0              | $\frac{1}{8}$ |
| 1        | 3   | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$  | $\frac{4}{8}$ |
| 2        | 3   | $\frac{3}{8}$ | $\frac{6}{8}$ | $\frac{12}{8}$ | $\frac{7}{8}$ |
| 3        | 1   | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{9}{8}$  | $\frac{8}{8}$ |
| $\Sigma$ | 8   | 1             | 1.5           | 3              |               |

← *cdf*

$$\text{Expected value (Mean)} = E(x) = \sum_{i=1}^n x_i P(x_i)$$

$$\therefore E(x) = 1.5$$

$$\text{variance} = E(x^2) - (E(x))^2$$

←  $x^2 P(x)$       ←  $(x P(x))^2$

$$V(x) = 3 - (1.5)^2$$

$$\therefore V(x) = 0.75$$

$$\text{Standard deviation} = \sqrt{V(x)}$$

$$\sigma = \sqrt{0.75}$$

$$\sigma = 0.866$$

## Ex 2 Two Dice

①  $x \Rightarrow$  Sum  $\rightarrow$  Two numbers

②  $x \Rightarrow$  Difference between Two numbers

Sol

①  $x \Rightarrow$  Sum  $\rightarrow$  Two numbers

|   | 1 | 2 | 3 | 4  | 5  | 6  |
|---|---|---|---|----|----|----|
| 1 | ② | 3 | 4 | 5  | 6  | 7  |
| 2 | 3 | 4 | 5 | 6  | 7  | 8  |
| 3 | 4 | 5 | 6 | 7  | 8  | 9  |
| 4 | 5 | 6 | 7 | 8  | 9  | 10 |
| 5 | 6 | 7 | 8 | 9  | 10 | 11 |
| 6 | 7 | 8 | 9 | 10 | 11 | ⑫  |

| $x$      | $f$ | $P(x)$         | $xP(x)$             | $x^2 P(x)$                                 | $F(x)$              |
|----------|-----|----------------|---------------------|--------------------------------------------|---------------------|
| 2        | 1   | $\frac{1}{36}$ | $\frac{2}{36}$      | $4 \times \frac{1}{36} = \frac{4}{36}$     | $\frac{1}{36}$      |
| 3        | 2   | $\frac{2}{36}$ | $\frac{6}{36}$      | $9 \times \frac{2}{36} = \frac{18}{36}$    | $\frac{3}{36}$      |
| 4        | 3   | $\frac{3}{36}$ | $\frac{12}{36}$     | $16 \times \frac{3}{36} = \frac{48}{36}$   | $\frac{6}{36}$      |
| 5        | 4   | $\frac{4}{36}$ | $\frac{20}{36}$     | $25 \times \frac{4}{36} = \frac{100}{36}$  | $\frac{10}{36}$     |
| 6        | 5   | $\frac{5}{36}$ | $\frac{30}{36}$     | $36 \times \frac{5}{36} = \frac{180}{36}$  | $\frac{15}{36}$     |
| 7        | 6   | $\frac{6}{36}$ | $\frac{42}{36}$     | $49 \times \frac{6}{36} = \frac{294}{36}$  | $\frac{21}{36}$     |
| 8        | 5   | $\frac{5}{36}$ | $\frac{40}{36}$     | $64 \times \frac{5}{36} = \frac{320}{36}$  | $\frac{26}{36}$     |
| 9        | 4   | $\frac{4}{36}$ | $\frac{36}{36} = 1$ | $81 \times \frac{4}{36} = \frac{324}{36}$  | $\frac{30}{36}$     |
| 10       | 3   | $\frac{3}{36}$ | $\frac{30}{36}$     | $100 \times \frac{3}{36} = \frac{300}{36}$ | $\frac{33}{36}$     |
| 11       | 2   | $\frac{2}{36}$ | $\frac{22}{36}$     | $121 \times \frac{2}{36} = \frac{242}{36}$ | $\frac{35}{36}$     |
| 12       | 1   | $\frac{1}{36}$ | $\frac{12}{36}$     | $144 \times \frac{1}{36} = \frac{144}{36}$ | $\frac{36}{36} = 1$ |
| $\Sigma$ | 36  | 1              | 7                   | $\frac{329}{6}$                            |                     |

$$E(x) = 7$$

$$V(x) = \frac{329}{6} - (7)^2$$

$$V(x) = 5.83$$

$$SD = \sqrt{5.83}$$

$$SD = 2.41$$



2  $x \rightarrow$  Difference between Two numbers

Die 2 1 2 3 4 5 6 Die 1

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 2 | 1 | 0 | 1 | 2 | 3 | 4 |
| 3 | 2 | 1 | 0 | 1 | 2 | 3 |
| 4 | 3 | 2 | 1 | 0 | 1 | 2 |
| 5 | 4 | 3 | 2 | 1 | 0 | 1 |
| 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| $x$      | $f$ | $P(x)$          | $xP(x)$         | $x^2 P(x)$      |
|----------|-----|-----------------|-----------------|-----------------|
| 0        | 6   | $\frac{6}{36}$  | 0               | 0               |
| 1        | 10  | $\frac{10}{36}$ | $\frac{10}{36}$ | $\frac{10}{36}$ |
| 2        | 8   | $\frac{8}{36}$  | $\frac{16}{36}$ | $\frac{32}{36}$ |
| 3        | 6   | $\frac{6}{36}$  | $\frac{18}{36}$ | $\frac{54}{36}$ |
| 4        | 4   | $\frac{4}{36}$  | $\frac{16}{36}$ | $\frac{64}{36}$ |
| 5        | 2   | $\frac{2}{36}$  | $\frac{10}{36}$ | $\frac{50}{36}$ |
| $\Sigma$ | 36  | 1               | $\frac{35}{18}$ | $\frac{35}{6}$  |

$$E(x) = \sum_{i=0}^{\infty} x_i P(x_i) = \frac{35}{18}$$

$$V(x) = E(x^2) - (E(x))^2$$

$$V(x) = \frac{35}{6} - \left(\frac{35}{18}\right)^2 = 2.1$$

$$SD = \sqrt{V(x)} = \sqrt{2.1} = 1.4$$

## → Continuous Random variables

Pdf  $\int_{-\infty}^{\infty} f(x) dx = 1$

Ex  $f(x) = 3x^2 \quad 0 \leq x \leq 1$

$$\int_0^1 3x^2 dx = 3 \cdot \frac{x^3}{3} \Big|_0^1 = x^3 \Big|_0^1 = 1 - 0 = 1$$

Cdf  $F(x) = \int_{-\infty}^x f(x) dx$

$$= \int_0^x 3x^2 dx = 3 \cdot \frac{x^3}{3} \Big|_0^x = x^3$$

$$F(x) = x^3$$

$f(x) = \frac{d}{dx} F(x)$   
→ Pdf      → Cdf

Mean  $\mu_x = \int_{-\infty}^{\infty} x f(x) dx$

Ex  $f(x) = 3x^2 \quad 0 \leq x \leq 1$

$$\int_0^1 x \cdot 3x^2 dx = 3 \int_0^1 x^3 dx = 3 \cdot \frac{x^4}{4} \Big|_0^1 = \frac{3}{4}$$



Variance  $\sigma^2 = \int_{-\infty}^{\infty} x^2 P(x) dx - \left( \int_{-\infty}^{\infty} x P(x) dx \right)^2$

$\xrightarrow{\text{Mean}}$

$$\sigma^2 = \int_0^1 x^2 \cdot 3x^2 dx - \left( \frac{3}{4} \right)^2$$

$$\sigma^2 = 3 \cdot \frac{x^5}{5} \Big|_0^1 - \frac{9}{16}$$

$$\sigma^2 = \frac{3}{5} - \frac{9}{16}$$

$$\sigma^2 = 0.04$$

Standard deviation

$$\sigma = \sqrt{\text{variance}}$$

$$\sigma = \sqrt{0.04} = 0.2$$

Ex

$$f(x) = \begin{cases} x & 0 \leq x \leq 1 \\ 2-x & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \int_0^2 f(x) dx$$

$$F(x) = \int_0^1 x dx + \int_1^2 2-x dx$$

$$F(x) = \frac{x^2}{2} \Big|_0^1 + \left( 2x - \frac{x^2}{2} \right) \Big|_1^2$$

$$F(x) = \frac{1}{2} + (4-2) - \left( 2 - \frac{1}{2} \right)$$

$$F(x) = 0.5 + 2 - 1.5$$

$$F(x) = 1$$

$\therefore$  Pdf